Práctica 2 — Modelado y control del sistema cardiovascular mediante el modelo Windkessel de cuatro elementos

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Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

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Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system('sistema2020_470P2_2')
parameters.StopTime = '15';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-2';
x = sim('sistema2020_470P2_2',parameters);

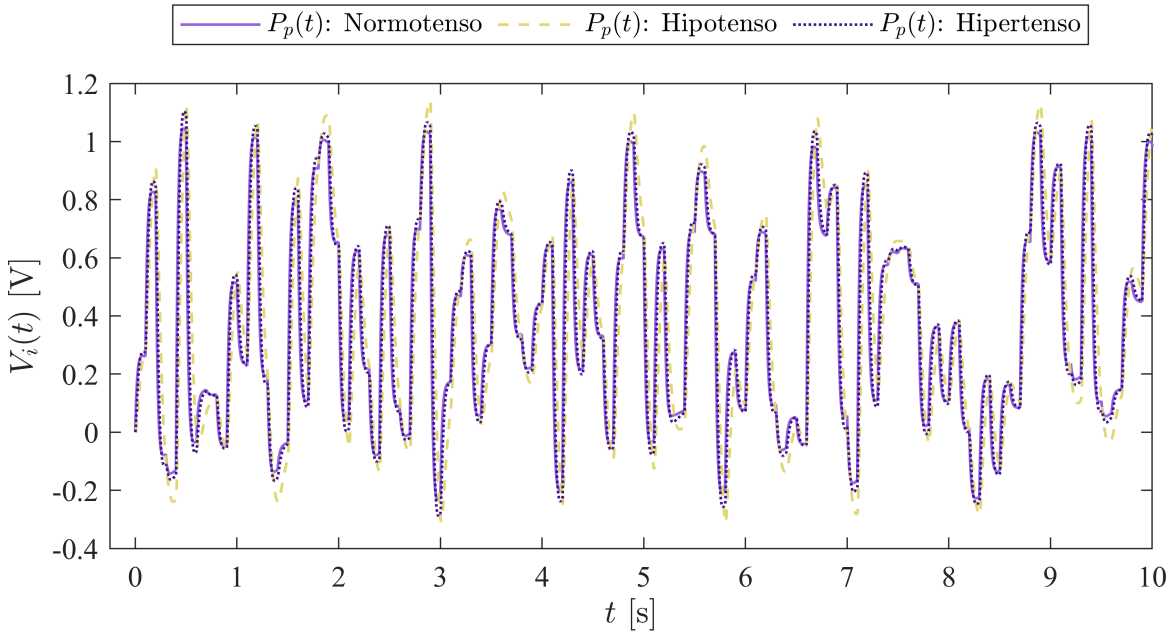
t_data = (0:0.1:15)';
t = (0:0.001:15)';
Pa = interp1(t_data, x.Pa, t, 'previous');
```

```
Pa = 15001×1
    0.2420
    0.2420
    0.2420
    0.2420
    0.2420
    0.2420
    0.2420
    0.2420
    0.2420
    0.2420
      ⋮
```

```
writematrix(x.Pa, 'signal.xlsx');
```

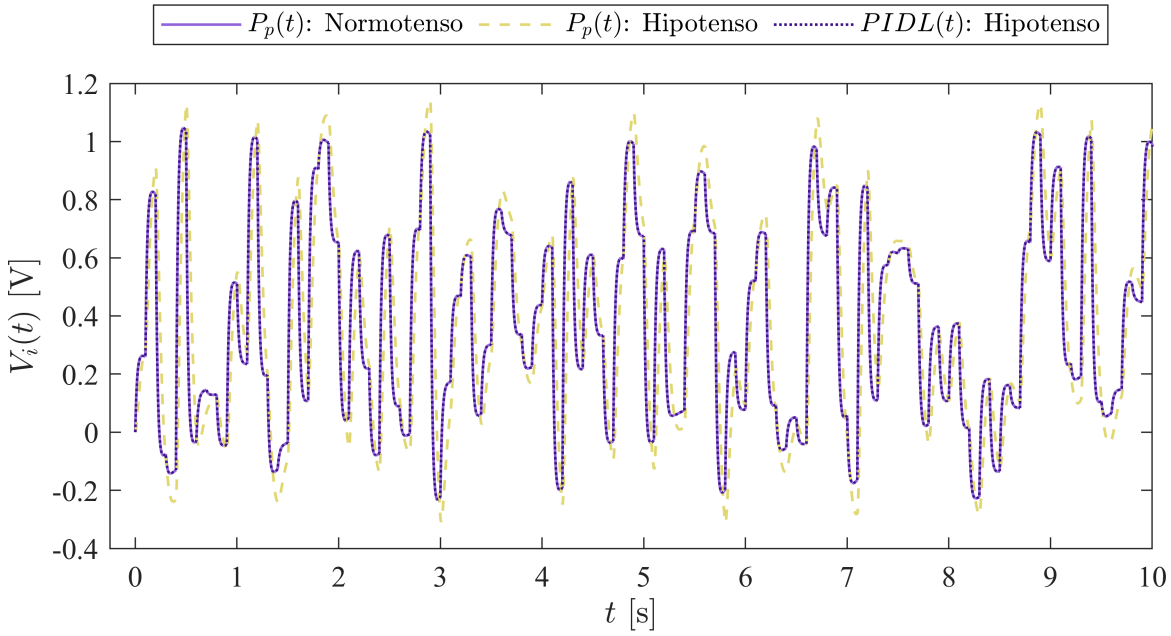
Respuesta en Individuos

```
plotsignals(x.t,x.PpN,x.PpL,x.PpH,'Individuos')
```



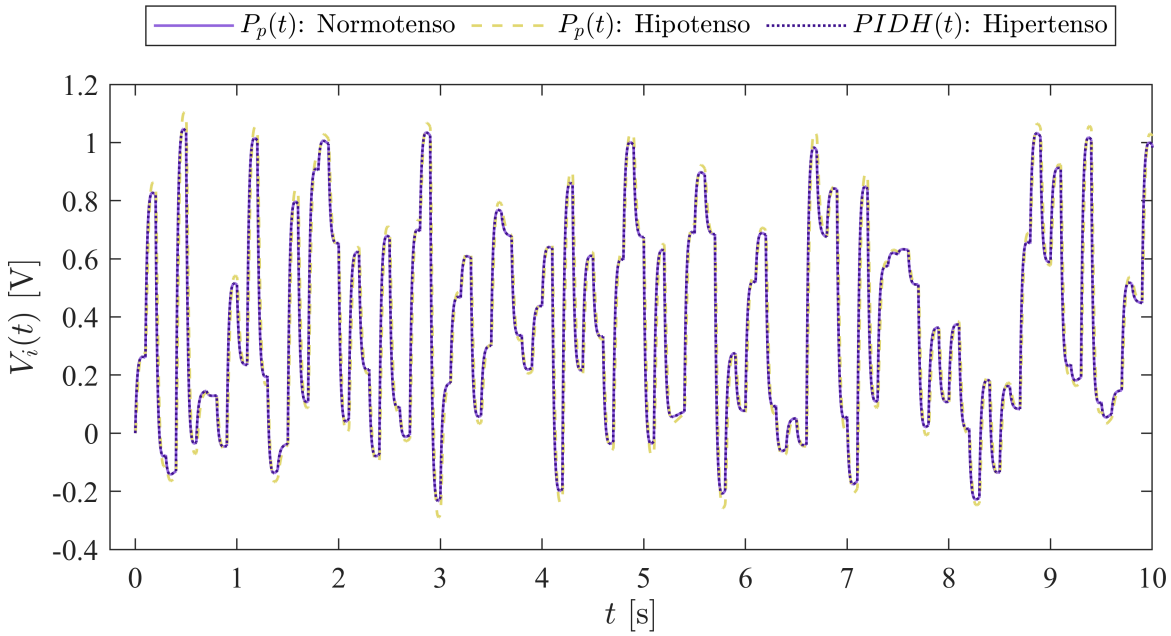
Respuesta: Normotenso, Hipotenso y Controlador

```
plotsignals(x.t,x.PpN,x.PpL,x.PIDL,'Hipotenso')
```



Respuesta: Normotenso, Hipertenso y Controlador

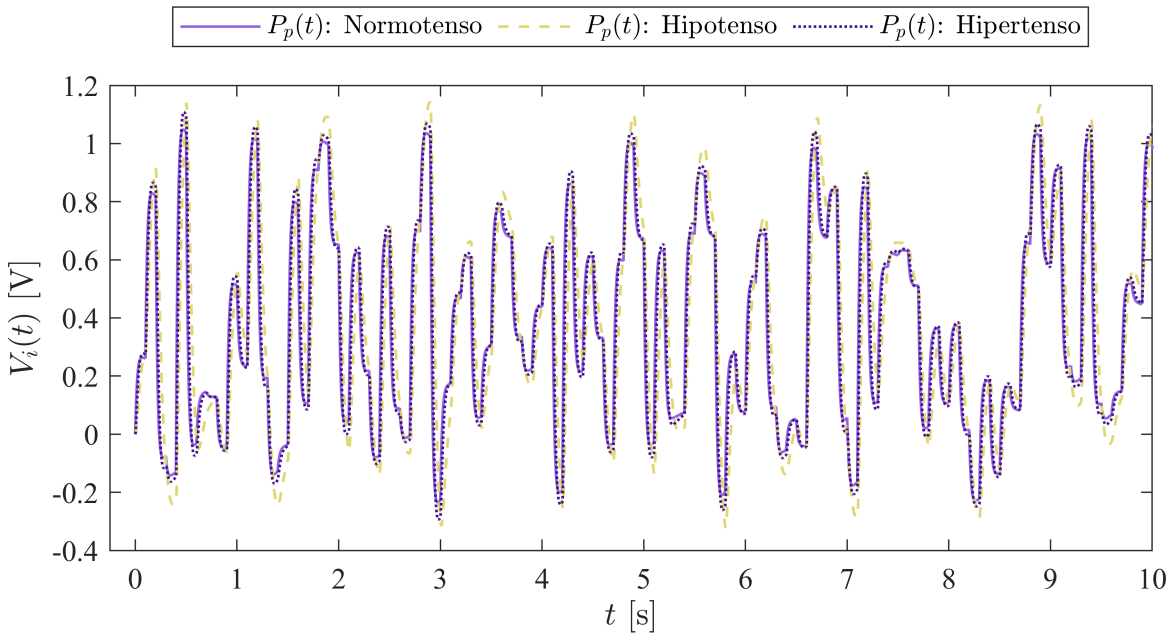
```
plotsignals(x.t,x.PpN,x.PpH,x.PIDH,'Hipertenso')
```



Modelo de EDOs

```
par_hipo = [0.01,0.001,1.00,3.500]; %[Z,L,R,C]
```

```
par_normo = [0.01,0.001,1.00,1.500];
par_hiper = [0.01,0.001,1.00,1.000];
PpN = cardio(par_normo,Pa,t);
PpL = cardio(par_hipo,PpN,t);
PpH = cardio(par_hiper,PpN,t);
plotsignals(t,PpN,PpL,PpH,'odes');
```




```
function Pp = cardio(par,Pa,t)
    Z = par(1); L = par(2); R = par(3); C = par(4); dt = t(2);
    n = round(t(end)/dt);
    FL = zeros(length(t),1); FL(1) = 0;
    Pp = zeros(length(t),1); Pp(1) = 0;
    for i = 1:n
        FL(i+1) = FL(i) + (-Pp(i)/L + Pa(i)/L)*dt;
        Pp(i+1) = Pp(i) + (FL(i)/C - (1/(Z*C) + 1/(R*C))*Pp(i) + Pa(i)/(Z*C))*dt;
    end
end
```

Funcion: Respuestas de la simulacion

```
function plotsignals (t,x1,x2,x3,signal)
    set(figure(),'color','w')
    set(gcf,'units','inches','position',[1,1,7,3.5])
    set(gca,'fontname','times new roman','fontsize',11)
    hold on; grid off; box on;
    clr1 = ([149, 100, 221])/256;
    clr2 = ([228, 218, 114])/256;
    clr3 = ([62, 15, 141])/256;
    order = [clr1;clr2;clr3]; colororder(order)
    xlabel('$t$ [s]','Interpreter','latex')
    ylabel('$V_i(t)$ [V]','Interpreter','latex')
    xlim([-0.25 10]); xticks(0:1:10)
```

```
    if signal == "Individuos"
        plot(t,x1,'-',t,x2,'--',t,x3,':','LineWidth',1)
        L = legend('$P_p(t)$: Normotenso','$P_p(t)$: Hipotenso','$P_p(t)$: Hipertenso');
    elseif signal == "Hipotenso"
        plot(t,x1,'-',t,x2,'--',t,x3,':','LineWidth',1)
        L = legend('$P_p(t)$: Normotenso','$P_p(t)$: Hipotenso','$PIDL(t)$: Hipotenso');
    elseif signal == "Hipertenso"
        plot(t,x1,'-',t,x2,'--',t,x3,':','LineWidth',1)
        L = legend('$P_p(t)$: Normotenso','$P_p(t)$: Hipotenso','$PIDH(t)$: Hipertenso');
    elseif signal == "odes"
                plot(t,x1,'-',t,x2,'--',t,x3,':','LineWidth',1)
        L = legend('$P_p(t)$: Normotenso','$P_p(t)$: Hipotenso','$P_p(t)$: Hipertenso');


    end

    set(L,'interpreter','latex','location','northoutside','orientation','horizontal')
    ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)

    exportgraphics(gcf,[signal,'_matlab.pdf'],'ContentType','vector')

end
```